PDN Noise and Decoupling Capacitors

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Introduction

This lab features a MOSFET slammer circuit driven by a square wave. The MOSFET pulls a significant amount of current from the power lines whenever the pulse wave sends a high signal. This circuit is to demonstrate the effect of switching on the voltage readings of the power line and the significance of decoupling capacitors to reduce the effect of noise on the power supply. This was accomplished by employing the following objectives.

1. Configure an Arduino to send a pulse wave to the MOSFET circuit
2. Ensure long interconnects between the power supply and MOSFET circuit
3. Calculate the inductance of the interconnects
4. Calculate the capacitance needed to mitigate the voltage drop on the power lines

Plan of Record

What does it mean to work?

1. The MOSFET must draw current in accordance with the pulse wave from the Arduino
2. The current going through board must stay within the current rating for all components
3. The power lines must experience a voltage drop with each pulse
4. The voltage drop must be limited with the introduction of the decoupling capacitor
5. The capacitor used for decoupling should have a capacitance consistent with prior calculations to reduce the voltage drop to 200 mV or below.

Features of the Board

1. Solderless breadboard with long interconnects between the power line and the MOSFET
2. Opamp to slow the rise time of the pulse wave and a switch to change between fast pulse (from Arduino) and slow pulse (from opamp)
3. 9v and 5v power lines, the 5v line will be used to power the components of the board and the 9v line will be directed to the collector pin of the MOSFET

Risk Reduction

1. Wires are color coded to reduce confusion when analyzing the circuit
2. Power wires from the Arduino are twisted to eliminate any noise induced by large loops
3. Wires are kept short to reduce large loop within the circuit
4. Average current through the components were calculated to ensure that components will not be damaged prior to turning the circuit on

Key Measurements

1. 5v and 9v coming from the respective power lines
2. Rise time of >10ns from Arduino
3. Rise time of ~10us from Opamp
4. Decrease of ~400mV for every pulse (or switch) from the MOSFET on the 9v power line
5. Average current >150 mA through the shunt resistor

Components:

1. TLV272 Opamp x1
2. TIP41C MOSFET x1
3. 10 Ohm resistor x1
4. Whadda Development Board (Arduino Equivalent) x1
5. 9v DC power plug x1

Constructing the Circuit:

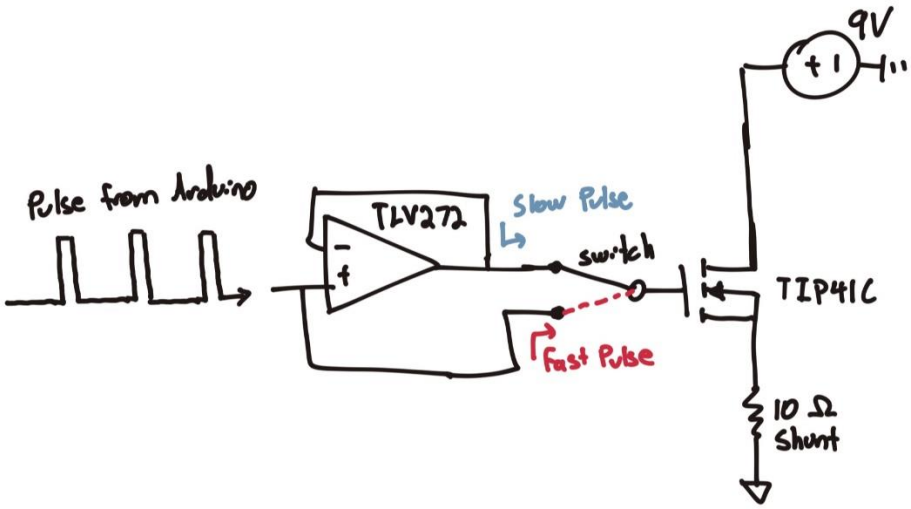


Fig 1: Napkin sketch of circuit

The circuit will be constructed according to figure 1, this configuration with the Opamp will allow for the testing of fast and slow signals and their effect on the power line. Before moving forward with the construction of the circuit, we must ensure the safety of the MOSFET and shunt resistors.

```
void setup() {
pinMode(13, OUTPUT);
}
void loop() {
digitalWrite(13, HIGH);
delay(1);
digitalWrite(13, LOW);
delay(20);
}
```

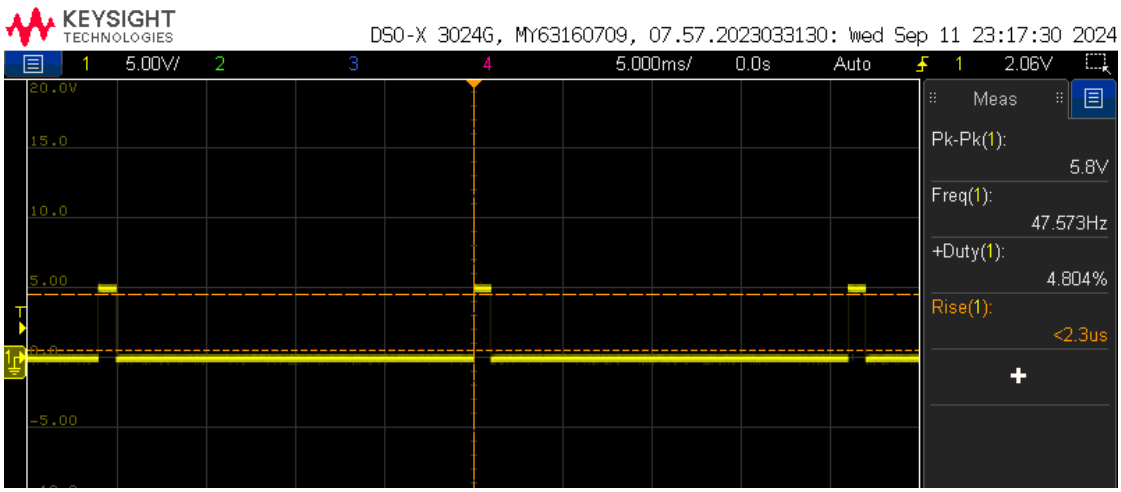


Fig 2: Code for Arduino (top) and resulting pulse wave (bottom)

The basis of these calculations will be the pulse wave from the Arduino from figure 2, the Arduino is outputting a pulse with a duty cycle of 4.8% and average voltage of .168 volts. The slow wave will show the same characteristics but with a slower rise time.

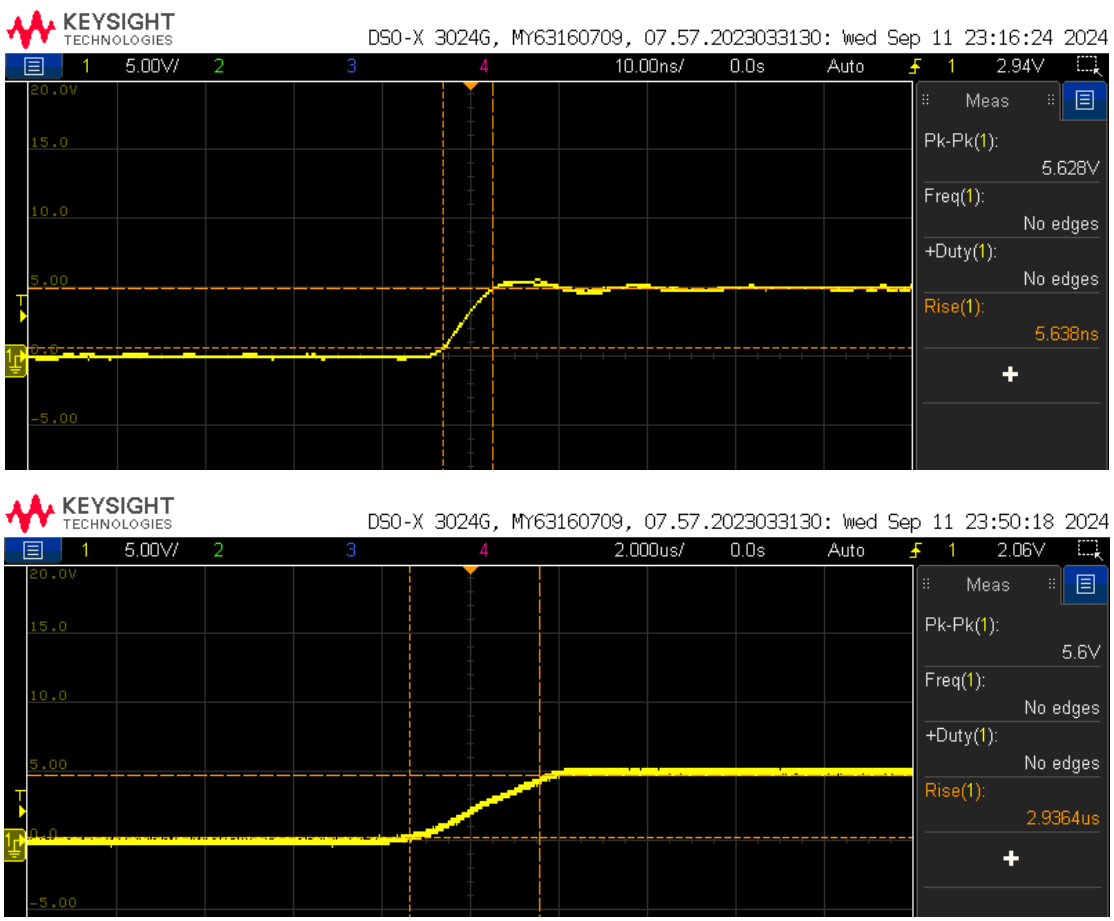


Fig 3: Rise time of fast pulse (top) and slow pulse (bottom)

Another metric that will be important for future calculations is the rise time of the pulse in both the fast and slow configurations. We find that the fast and slow pulses have a rise time of 5.64 ns and 2.94 us respectively.

Calculations:

Checking for the safety of the MOSFET and resistor, we must ensure that the components never experience a current higher than what it is rated for. According to the data sheet, the TIP41C is rated for 3 amps across the base and emitter pins and the resistor is rated for a power dissipation of .25 W.

Expected current @ shunt:

$V_{Thresh} = 1.5V$ (from datasheet)

$V_{in} = 5.8V$ (when pin is HIGH)

$R_{shunt} = 10\Omega$

$$\frac{V_{in} - V_{Thresh}}{R_{shunt}} = \frac{5.8 - 1.5}{10} = \frac{4.3}{10} = 430\,mA \checkmark$$

*Safe for MOSFET

Duty cycle = 4.8%

$\therefore V_{avg} = 4.3\,(.048) = .2064$

$$I_R = \frac{.2064}{10} = .02064\,A$$

$$P_R = (.02064)^2(10) = .00426\,W$$

$.00426\,W < .25\,W \checkmark$

*Safe for Resistor

Fig 4: Calculation for the current going through the MOSFET(left) and power dissipation of the resistor (right)

Since the shunt resistor is the current limiting factor, the current through the shunt is the same current going through the MOSFET's base and emitter pins. Using this, we calculated that 430 mA is passing through the MOSFET for every high signal.

The average power dissipated through the shunt was found to be .00426 W.

The current passing through the circuit is well within the ratings of the components, therefore damage to the components is not a concern.

After calculating the current through the circuit, the inductance on the long interconnects were calculated to understand the voltage drops occurring on the power line.

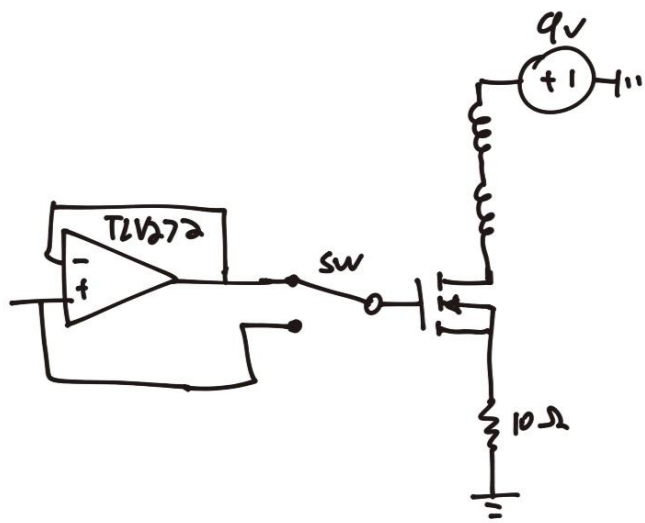


Fig 5: Equivalent circuit with loop inductance on power line.

According to figure 5, the long interconnects on the 9v power line can be seen as a series of inductors, with the voltage drop across the inductors intensifying the further the measurement is taken from the ICs. Recognizing that the voltage drop across an inductor is given by the following equation

$$\Delta V = L\left(\frac{dI}{dt}\right)$$

We can expect there to be a larger change in voltage with fast changes in current and smaller changes in voltage with slow changes in current.

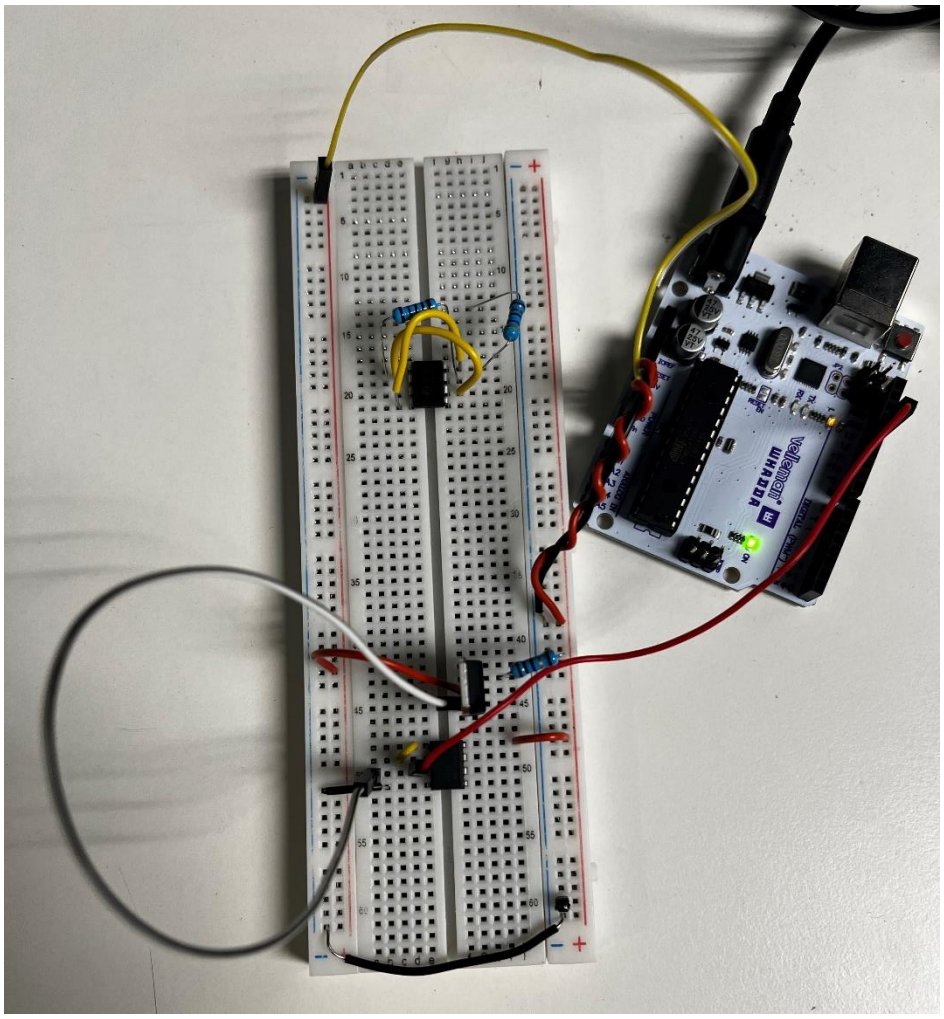


Fig 6: SBB version of napkin sketch circuit

NOTE: the resistors and timer on the top half of the SBB are from another lab and do not pertain to this report.

The 9v rail is located on the left side of the SBB. Additionally, the switch will be the white and black jumper wire instead of a physical switch. The wire will connect to either the opamp output or Arduino output depending on the speed of switching we want.

NOTE: All scope screenshots using different scaling for scope 1 (yellow) and 2 (green). As scope 2 is the signal on the power line, the screenshots will show the scaling for scope 2 since that information pertains to the goal of this lab.

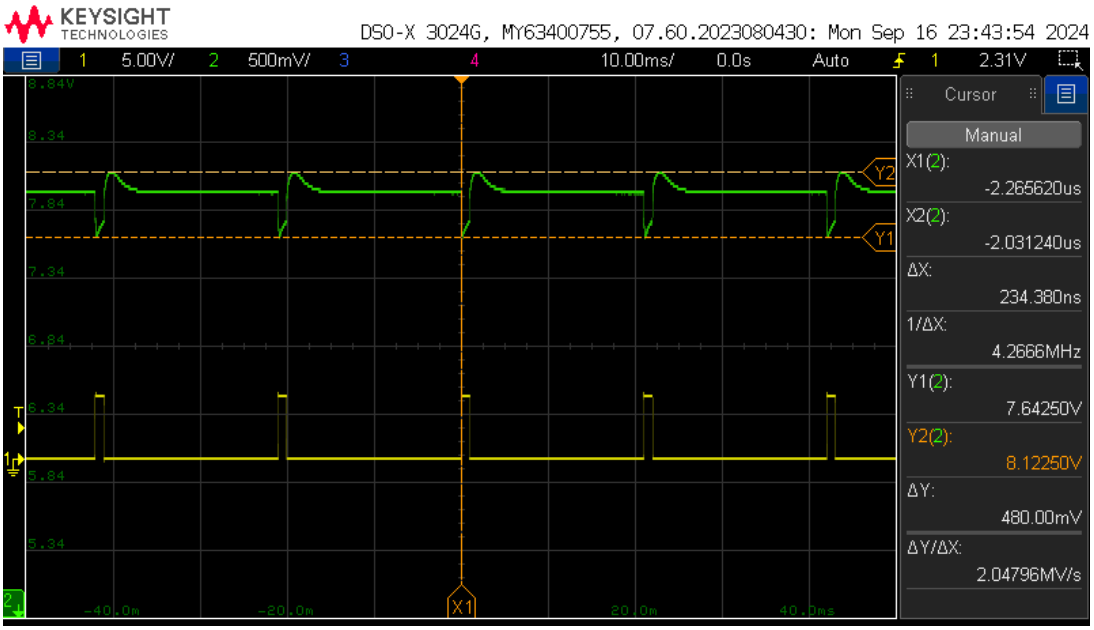
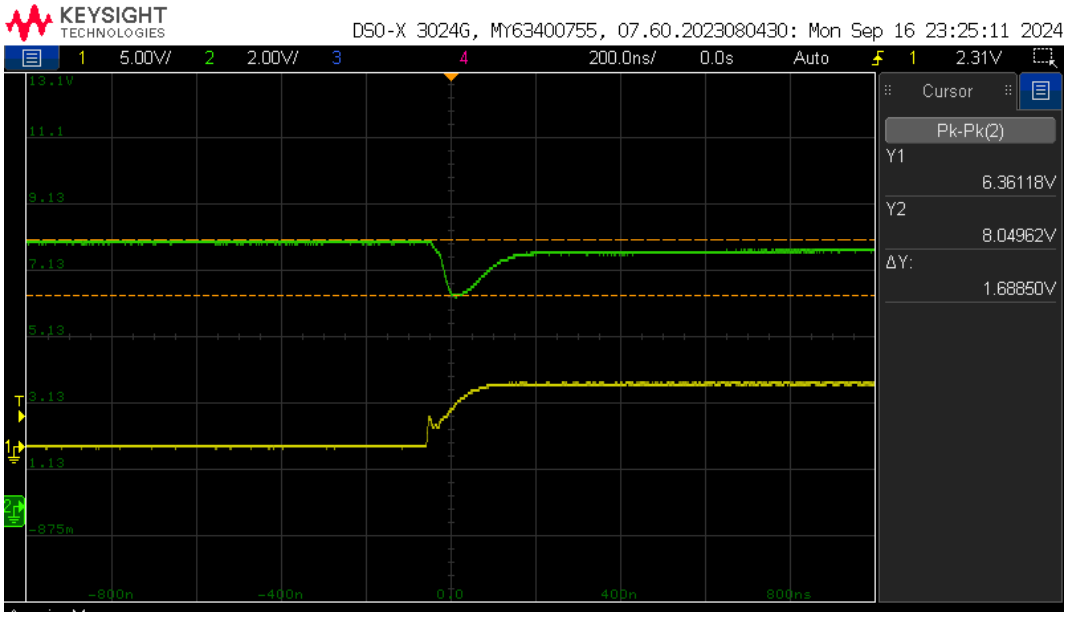


Fig 7: Signature on 9v line

Before analyzing the switching noise, the overall effect of the switching on the power line should be observed. From figure 7, we see that there is significant drops in voltage every time the Arduino switches, this is enough of a drop to warrant concern for the functionality of other ICs involved in the circuit.



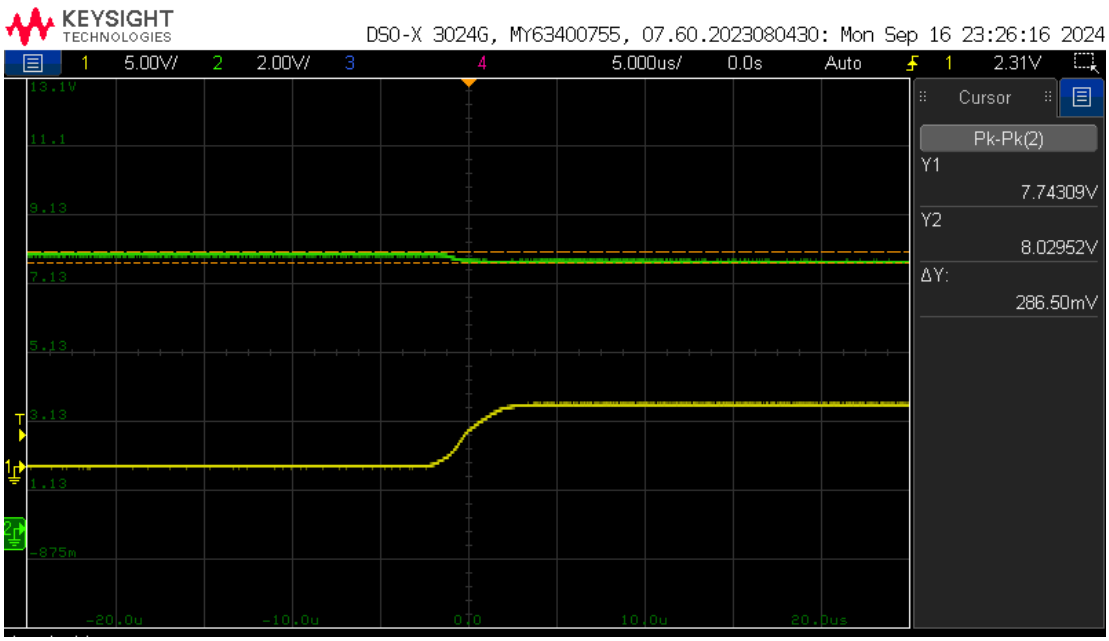


Fig 8: Switch noise from the fast (top) and slow (bottom) pulses

NOTE: The time over which noise occurs were left out of the measurements by mistake, for future reference, the dt on the fast-switching noise was measured to be 234 ns.

As is evident in figure 6, there is a larger voltage drop (delta Y) when the circuit is in the fast pulse configuration compared to the slow pulse. From this, the inductance of the interconnects can be calculated, this information will be useful when constructing other circuits using this breadboard.

We get a ΔV of 1.4V over 234ns w/ fast switch...

$$\therefore 1.4 = L\left(\frac{.430}{234\,ns}\right)$$

$$L = 748\ nH$$

Fig 9: Calculated inductance of interconnects

From the above calculation, the interconnect has an inductance of 748 nH.

Sudden voltage drops on the power line can interfere with the function of other ICs, so it is within our best interests to negate this effect as much as possible. One possible way to do that is to introduce a decoupling capacitor. While it is possible to use the largest possible capacitor to reduce noise, using enough of these may cut into the budget of the board. Therefore, it would better to calculate the capacitance needed to achieve certain reduction in voltage drop.

The equation for the voltage drops in terms of capacitance is as follows:

$$\frac{dV}{dt} = \frac{1}{C} * i$$

Using this equation to keep the drop in voltage to under .4 V…

Finding necessary C for decoup capacitor:

Recall: $\frac{dV}{dt} = \frac{1}{C}(I)$

if $I = 430$ mA…

$\frac{dV}{dt} = \frac{1}{C}(.43)$

keep below .4v …

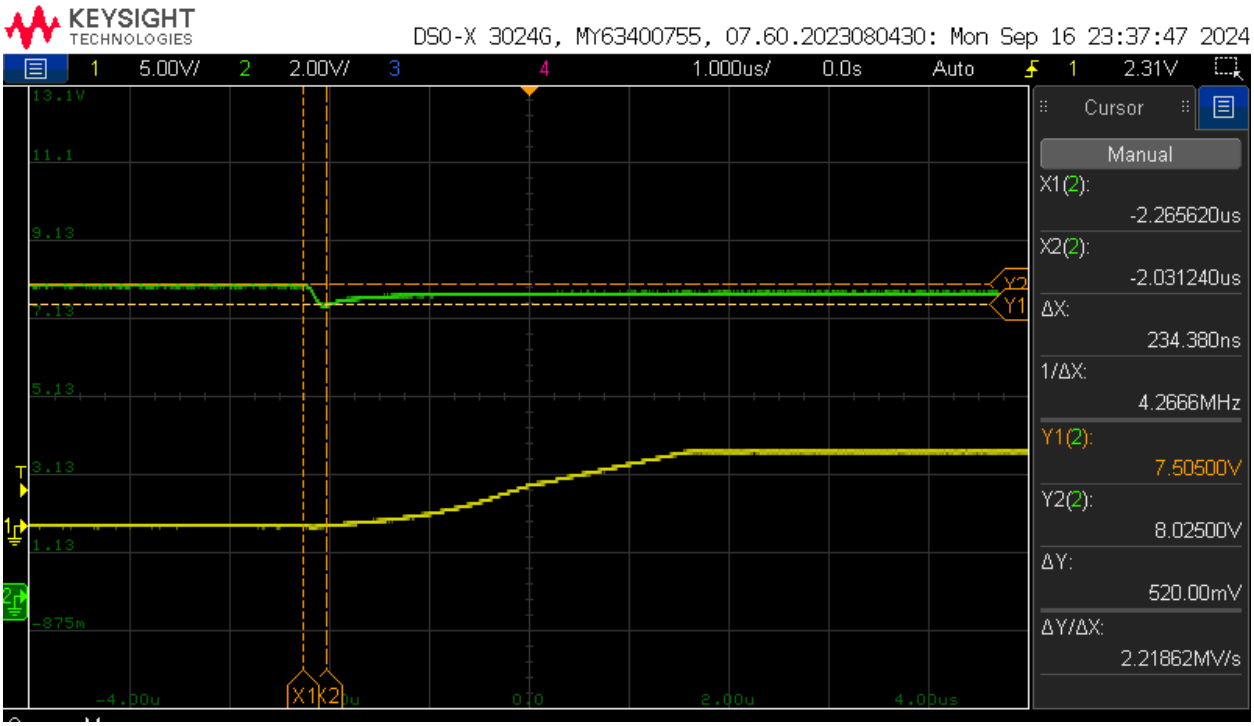
$\frac{.4}{234 \times 10^{-9}} = \frac{1}{C}(.43)$ => $C \approx 1.2\,\mu F$

use $C = 1\,\mu F$

Fig 10: Calculation for minimum needed capacitance

Decoupling Capacitor Usage and Placement

From figure 10, we would need a 1.2 uF capacitor. The closest capacitor available is 1 uF, therefore, the 1 uF capacitor will be used. To demonstrate the effect of capacitance, both the 1- and 1000 uF capacitors will be used. The smaller capacitor reduces the voltage drop to a little higher than 400 mV (since we are using a smaller capacitor than what was calculated) whereas the larger should reduce the voltage drop far below 400 mV. Placing these capacitors close to the ICs results in the following signature on the powerline.



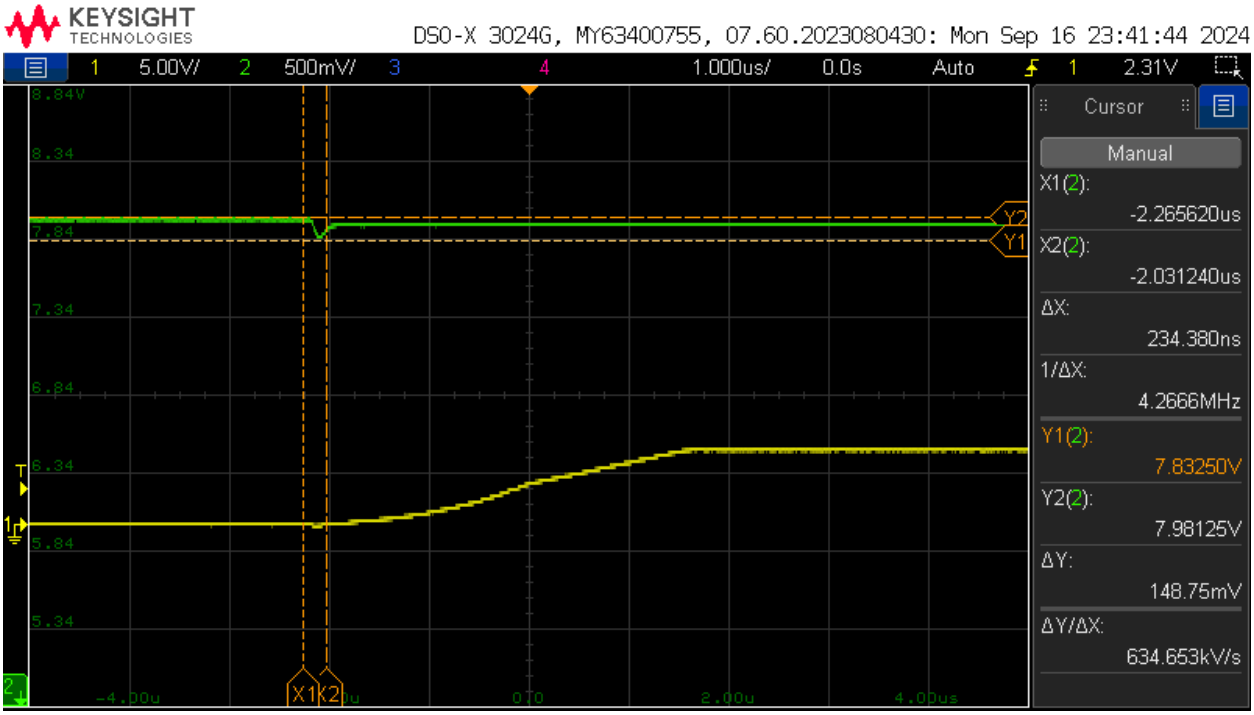
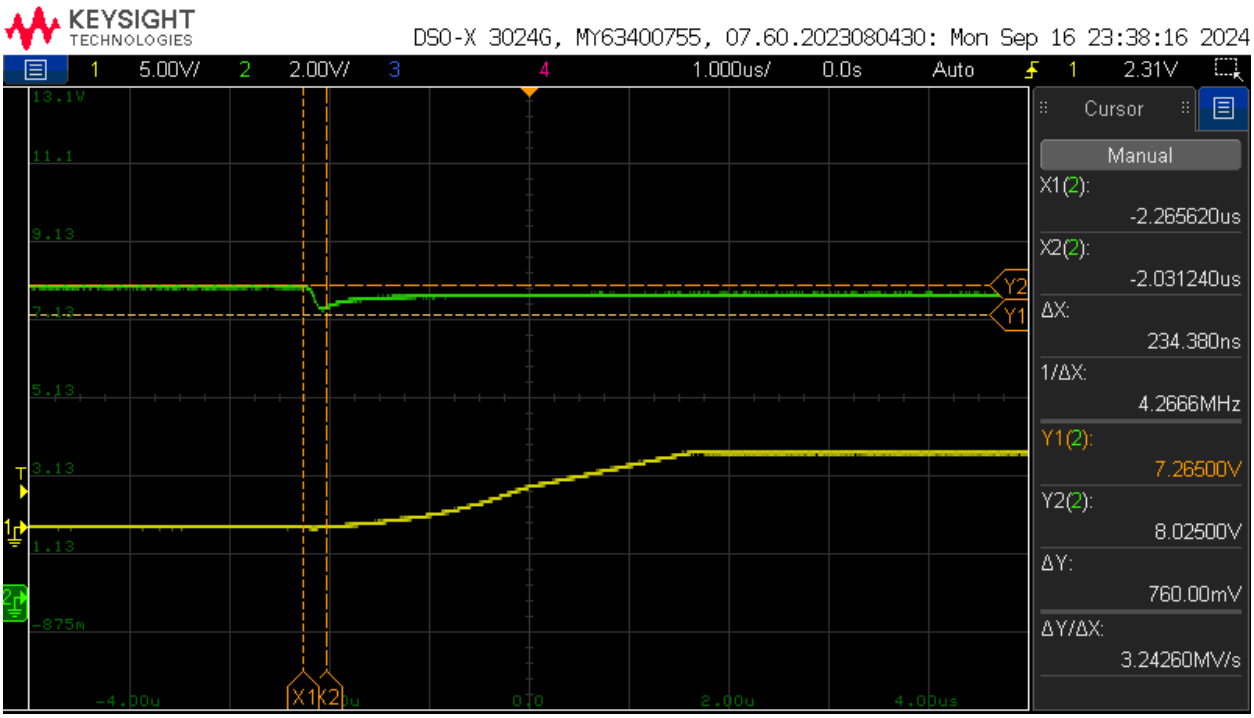


Fig 11: Noise on the power lines after introducing 1 uF capacitor (top) and 1000 uF capacitor (bottom)

Evidently, both capacitors show a reduction of noise, the smaller of the two reducing the voltage drop to 520 mV which is consistent with the calculations in figure 10. The larger capacitor can reduce the voltage drop to about 148 mV.



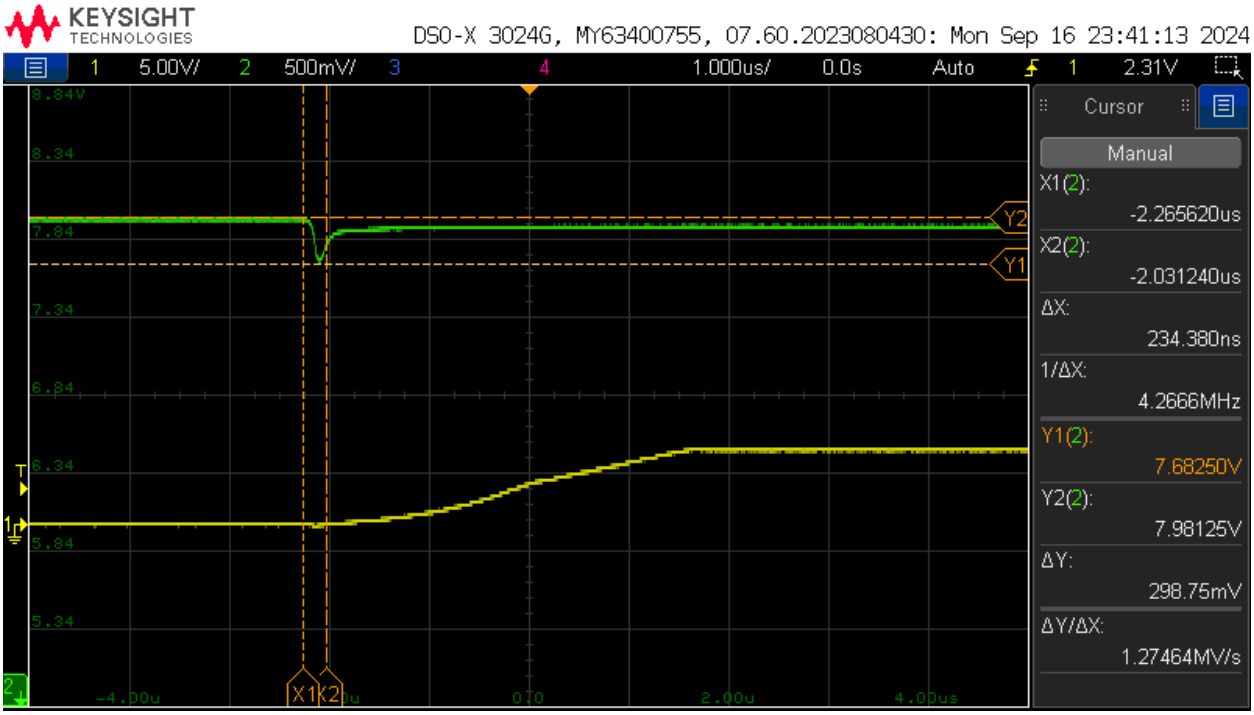


Figure 10: Placing the 1uF (top) and 1000uF (bottom) decoupling capacitors far from the ICs

Another consideration is the placement of the decoupling capacitors. Placing them farther away from the ICs results in a smaller reduction in noise. Therefore, it is best practice to place decoupling capacitors as close as possible to the ICs.

Conclusions and Key Takeaways:

The purpose of this lab was to understand the effects of long interconnects in circuits and why this setup is not ideal for circuits dealing with high switching speeds. Long interconnects behave like inductors, and since inductors opposes change in current, fast switching (which is effectively a fast change in current) will induce a voltage drop on that line. To reduce this effect so that all other ICs function as normal, we can introduce a decoupling capacitor. According to the plan of record, this circuit working and was effective in demonstrating the effects of long interconnects and decoupling capacitors. The biggest takeaways from this lab were the following.

1. Decoupling capacitors should be used in every circuit, especially those with high switching speeds. This ensures the safety of the ICs as well as consistent function.

2. Decoupling capacitors should be placed as close as possible to ICs. This way most of the noise in major signal traces can be limited.

3. Observing the switching speed and resulting noise prior to building the final circuit should be done to identify the major sources of noise. Knowing this beforehand will make it easier to troubleshoot in the future.

4. Calculating the inductance of the interconnects should be done so that any unwanted noise can be anticipated and addressed.

5. The change in voltage for every switch should be noted and used to calculate the minimum required capacitance as opposed to using the maximum capacitor available. This is to avoid allocating excess resources for noise reduction.